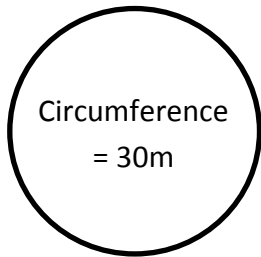


Red Example:



$$\text{Diameter} = \text{circumference} \div \pi$$

$$\text{Diameter} = 30 \div \pi$$

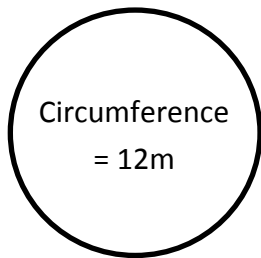
$$\text{Diameter} = 9.549296586\text{m}$$

$$\text{Diameter} = 9.55\text{m}$$

$$\text{Radius} = 9.55\text{m} \div 2$$

$$\text{Radius} = 4.78\text{m}$$

1)



$$\text{Diameter} = \text{circumference} \div \pi$$

$$\text{Diameter} = 12 \div \pi$$

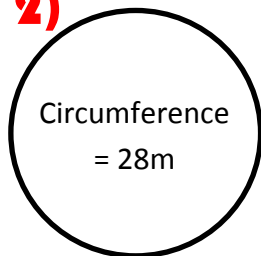
$$\text{Diameter} = 3.819718634\text{m}$$

$$\text{Diameter} = \underline{\hspace{2cm}}\text{m}$$

$$\text{Radius} = \underline{\hspace{2cm}}\text{m} \div 2$$

$$\text{Radius} = \underline{\hspace{2cm}}\text{m}$$

2)



$$\text{Diameter} = \underline{\hspace{2cm}} \div \pi$$

$$\text{Diameter} = \underline{\hspace{2cm}}\text{m}$$

$$\text{Diameter} = \underline{\hspace{2cm}}\text{m}$$

$$\text{Radius} = \underline{\hspace{2cm}}\text{m} \div 2$$

$$\text{Radius} = \underline{\hspace{2cm}}\text{m}$$

Amber

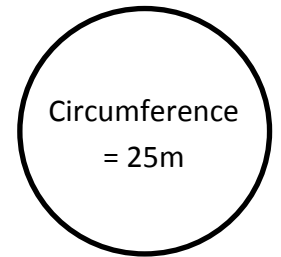
1) $\text{Diameter} = \underline{\hspace{2cm}} \div \pi$

$$\text{Diameter} = \underline{\hspace{2cm}}\text{m}$$

$$\text{Diameter} = \underline{\hspace{2cm}}\text{m}$$

$$\text{Radius} = \underline{\hspace{2cm}}\text{m} \div 2$$

$$\text{Radius} = \underline{\hspace{2cm}}\text{m}$$



2) $\text{Diameter} = \text{circumference} \div \pi$

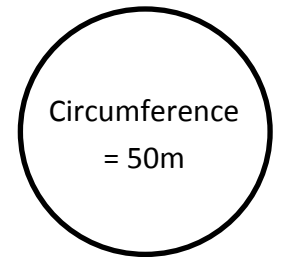
$$\text{Diameter} =$$

$$\text{Diameter} =$$

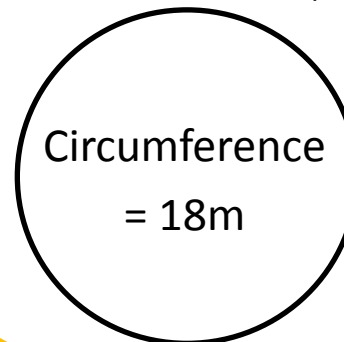
$$\text{Diameter} =$$

$$\text{Radius} =$$

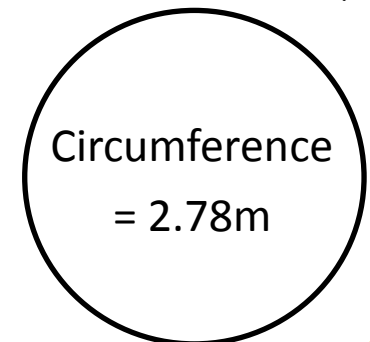
$$\text{Radius} =$$



3) Calculate, to 2dp, the diameter and radius of:



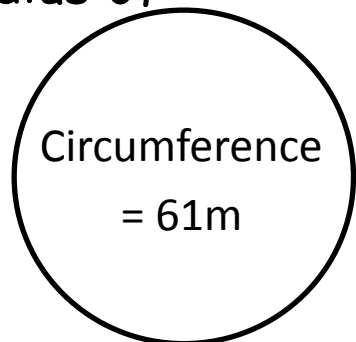
4) Calculate, to 2dp, the diameter and radius of:



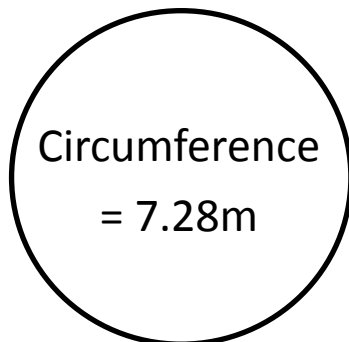
Green

Calculate, to 2dp, the diameter and radius of:

1)



2)



3) $C = 8\text{m}$

8) $C = 33.3\text{m}$

4) $C = 16\text{m}$

9) $C = 6.28\text{m}$

5) $C = 24\text{m}$

10) $C = 10.55\text{m}$

6) $C = 64\text{m}$

11) $C = 1024\pi\text{m}$

7) $C = 20\text{m}$

12) $C = 5\sqrt{2}\text{m}$

Extension

- 1) A square has a side length of 8cm, what would be the radius of a circle with the same perimeter to 2dp?



Draw a diagram first!

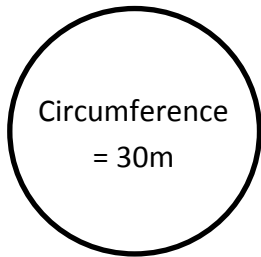
- 2) A team of adventurers are planning a journey on a mysterious island, which is, astonishingly, perfectly circular.

One team is planning on doing a lap around the edge of the island and another is travelling straight through the centre and back. The teams travel at the same speed and arrive back at the same time. What are the possible dimensions of the island?



Try a couple of lengths, for example: if the island was 1km in diameter, how long would each journey be?

Red Example:



$$\text{Diameter} = \text{circumference} \div \pi$$

$$\text{Diameter} = 30 \div \pi$$

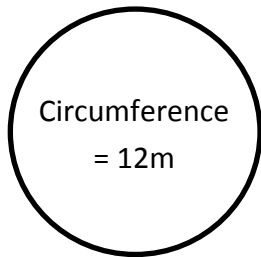
$$\text{Diameter} = 9.549296586\text{m}$$

$$\text{Diameter} = 9.55\text{m}$$

$$\text{Radius} = 9.55\text{m} \div 2$$

$$\text{Radius} = 4.78\text{m}$$

1)



$$\text{Diameter} = \text{circumference} \div \pi$$

$$\text{Diameter} = 12 \div \pi$$

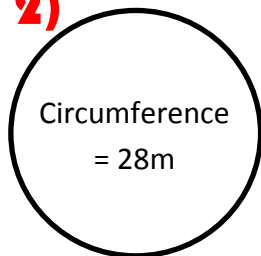
$$\text{Diameter} = 3.819718634\text{m}$$

$$\text{Diameter} = \underline{\hspace{2cm}}\text{m}$$

$$\text{Radius} = \underline{\hspace{2cm}}\text{m} \div 2$$

$$\text{Radius} = \underline{\hspace{2cm}}\text{m}$$

2)



$$\text{Diameter} = \underline{\hspace{2cm}} \div \pi$$

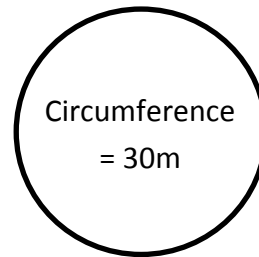
$$\text{Diameter} = \underline{\hspace{3cm}}\text{m}$$

$$\text{Diameter} = \underline{\hspace{2cm}}\text{m}$$

$$\text{Radius} = \underline{\hspace{2cm}}\text{m} \div 2$$

$$\text{Radius} = \underline{\hspace{2cm}}\text{m}$$

Red Example:



$$\text{Diameter} = \text{circumference} \div \pi$$

$$\text{Diameter} = 30 \div \pi$$

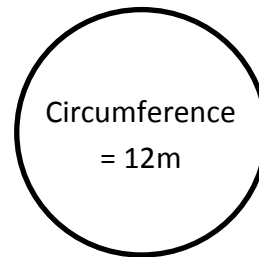
$$\text{Diameter} = 9.549296586\text{m}$$

$$\text{Diameter} = 9.55\text{m}$$

$$\text{Radius} = 9.55\text{m} \div 2$$

$$\text{Radius} = 4.78\text{m}$$

1)



$$\text{Diameter} = \text{circumference} \div \pi$$

$$\text{Diameter} = 12 \div \pi$$

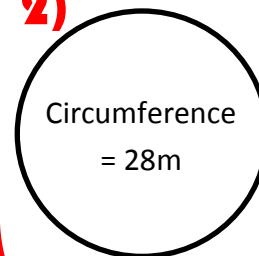
$$\text{Diameter} = 3.819718634\text{m}$$

$$\text{Diameter} = \underline{\hspace{2cm}}\text{m}$$

$$\text{Radius} = \underline{\hspace{2cm}}\text{m} \div 2$$

$$\text{Radius} = \underline{\hspace{2cm}}\text{m}$$

2)



$$\text{Diameter} = \underline{\hspace{2cm}} \div \pi$$

$$\text{Diameter} = \underline{\hspace{3cm}}\text{m}$$

$$\text{Diameter} = \underline{\hspace{2cm}}\text{m}$$

$$\text{Radius} = \underline{\hspace{2cm}}\text{m} \div 2$$

$$\text{Radius} = \underline{\hspace{2cm}}\text{m}$$

Amber

1) Diameter = ____ $\div \pi$

Diameter = _____m

Diameter = _____m

Radius = _____m $\div 2$

Radius = _____m

Circumference
= 25m

2) Diameter = circumference $\div \pi$

Diameter =

Diameter =

Diameter =

Radius =

Radius =

Circumference
= 50m

3) Calculate, to 2dp, the diameter and radius of:

Circumference
= 18m

4) Calculate, to 2dp, the diameter and radius of:

Circumference
= 2.78m

Amber

1) Diameter = ____ $\div \pi$

Diameter = _____m

Diameter = _____m

Radius = _____m $\div 2$

Radius = _____m

Circumference
= 25m

2) Diameter = circumference $\div \pi$

Diameter =

Diameter =

Diameter =

Radius =

Radius =

Circumference
= 50m

3) Calculate, to 2dp, the diameter and radius of:

Circumference
= 18m

4) Calculate, to 2dp, the diameter and radius of:

Circumference
= 2.78m

Answers Red:

1) $d = 3.82\text{m}$, $r = 1.91\text{m}$ 2) $d = 8.91\text{m}$, $r = 4.46\text{m}$

Amber:

1) $d = 7.96\text{m}$, $r = 3.98\text{m}$ 2) $d = 15.92\text{m}$, $r = 7.96\text{m}$ 3) $d = 5.73\text{m}$, $r = 2.87\text{m}$
4) $d = 0.88\text{m}$, $r = 0.44\text{m}$

Green:

1) $d = 19.42\text{m}$, $r = 9.71\text{m}$ 2) $d = 2.32\text{m}$, $r = 1.16\text{m}$ 3) $d = 2.55\text{m}$, $r = 1.28\text{m}$
4) $d = 5.09\text{m}$, $r = 2.55\text{m}$ 5) $d = 7.64\text{m}$, $r = 3.82\text{m}$ 6) $d = 20.37\text{m}$, $r = 10.19\text{m}$
7) $d = 6.37\text{m}$, $r = 3.19\text{m}$ 8) $d = 10.6\text{m}$, $r = 5.3\text{m}$ 9) $d = 2\text{m}$, $r = 1\text{m}$
10) $d = 3.36\text{m}$, $r = 1.68\text{m}$ 11) $d = 1024\text{m}$, $r = 512\text{m}$ 12) $d = 2.25\text{m}$, $r = 1.13\text{m}$

Extension:

1) 5.1m 2) It can't be done, one team travels πd and other travels $2d$